Neonatal Hypothermia

67

Background

- ☑ Thermoregulation is the ability to maintain balance between heat production and heat loss in order to sustain body temperature within a normal range.
- ☑ Hypothermia may have serious metabolic consequences for all newborns. To minimize these effects, an environmental temperature at which the newborn has minimal rates of oxygen consumption and expends the least energy to maintain its temperature is needed. This is known as thermoneutral environment.

Definition

The World Health Organization (WHO) defines neonatal hypothermia as an axillary temperature below 36.5°C (97.7°F) among newborns aged below 28 days. Normal axillary temperature is 36.5–37.5°C.

- ☑ Mild hypothermia/cold stress 36.0–36.4°C
- ☑ Moderate hypothermia 32.0–35.9°C
- ☑ Severe hypothermia <32°C.

Risk Factors

- ☑ Prematurity, low birth weight, intrauterine growth restriction (IUGR), asphyxia, and congenital defects like abdominal wall defects.
- ☑ Low delivery room temperature, not drying the neonate during postdelivery care, bathing the newborn after birth, removal of vernix caseosa, reduced contact with mother, and delayed initiation of breastfeeding are risk factors for neonatal hypothermia, especially in developing countries.
- ☑ Poor understanding of healthcare providers about the physiology of thermoregulation is a contributory factor too.
- ☑ Low ambient temperature as well as lower maternal temperature are known risk factors. Neonatal transport is almost always done poorly and is a risk factor for hypothermia.
- ☑ Procedures for neonatal care such as surgery, placement of umbilical lines, and radiological investigations such as MRI are associated with neonatal hypothermia.

Prevalence

Across the gestational age spectrum of newborns that are cared for in various settings (including warm tropical climates), the prevalence of hypothermia varies extensively from 8 to 92% across various studies. In developing countries, the typical rates are 32–85% in hospital settings.

Outcomes of Neonatal Hypothermia

- ☑ Neonatal hypothermia is an independent risk factor for neonatal mortality across all gestational ages with association showing a dose-response relationship.
- ☑ Mortality increases by 28% per 1°C decrease in temperature below the normal temperature.
- ☑ Neonatal hypothermia on admission to neonatal intensive care unit (NICU) has also been consistently associated with intraventricular hemorrhage, bronchopulmonary dysplasia, neonatal sepsis, retinopathy of prematurity, and increased length of hospital stay.

Measurement of Temperature

- ☑ Axilla is the recommended site of measurement using a digital thermometer. The thermometer is placed in the baby's armpit and the arm is held close to the body to keep it in place for about 15 seconds or till it beeps.
- ☑ The temperature is displayed on the thermometer. Temperature can also be recorded continuously by a thermistor attached to a radiant warmer or incubator with the probe attached to the skin over the upper abdomen.
- ☑ The thermistor senses the skin temperature and displays it on the panel. Both the above methods are acceptable. Rectal measurement is not preferred.
- ☑ Noninvasive measurements of neonatal temperature using infrared thermometers or infrared thermography are not recommended as they are not yet reliable. Novel bracelet devices placed on newborn's wrists to detect hypothermia are fairly accurate detectors of hypothermia and may be used in appropriate settings.

Mechanism of Heat Production in Newborns

- ☑ *Nonshivering thermogenesis*—occurs by utilizing brown fat in newborns. Thermoreceptors on sensing a low temperature result in elevated sympathetic output and this stimulates the beta-adrenergic receptors in the brown fat increasing cAMP. This results in increased metabolism and increases heat production.
- ☑ *Increased metabolic activity*—the brain, heart, and liver produce metabolic energy by oxidative metabolism of glucose, fat, and protein.
- ☑ *Peripheral vasoconstriction*—reduces blood flow to the skin and decreases loss of heat.

Mechanisms of Heat Loss

TABLE 1: Mechanisms of heat loss.	
Evaporation	Due to the evaporation of amniotic fluid from skin surface
Conduction	By coming in contact with cold objects such as cloth and weighing tray
Convection	Convection by air currents where cold air replaces warm air around baby due to open windows, fans, etc.
Radiation	Radiation to colder solid objects in vicinity-like walls

The process of heat gain is by conduction, convection, and radiation.

Clinical Features

TABLE 2: Clinical features of neonatal hypothermia.	
Peripheral vasoconstriction	Acrocyanosis, cool/pale extremities, and decreased peripheral perfusion
Central nervous system (CNS) depression	Lethargy, hypotonia, bradycardia, apnea, and poor feeding
Increased metabolism	Hypoglycemia, hypoxia, and metabolic acidosis
Increased pulmonary artery pressure	Respiratory distress and tachypnea
Chronic signs	Disseminated intravascular coagulation (DIC) and poor weight gain

☑ In delivery room and operation theater:

- Follow the 10 steps of "warm chain" recommended by the WHO.
  - Draught free and warm delivery room temperature of 25–28°C.
  - Radiant warmer to be prewarmed along with all the linen and clothes/cap before delivery.
  - Cap prevents significant heat loss in preterm as well as in term infants. Remove wet towel.
  - Baby is placed directly on the mother's abdomen or chest after delivery in both vaginal and cesarean delivery.
  - Provide warmth by skin-to-skin contact after drying with a warm and dry linen if baby is doing well.
  - Breastfeeding can be started immediately and the baby and the mother are covered with a warm blanket. Delay bathing. No bathing in the hospital.
  - Resuscitation, if required, should be done under the radiant warmer and heated humidified gases to be used if oxygen or positive pressure ventilation is required.
  - Prewarm medications and intravenous (IV) fluid, if required.
  - During surgery, abdominal organ coverage reduces the incidence of hypothermia.
  - Additional measures for very preterm infants (who are more prone to hypothermia due to greater surface-to-mass ratio and lesser brown fat):

<28 weeks gestational age (GA)	***<32 weeks GA***
Plastic wrap covered up to the neck (without drying) along with Transwarmer mattresses is preferred.	Plastic wraps covered up to the neck without drying.
Head is dried and cap placed.	
Transport and management in a heated humidified incubator reduces heat and water loss.	

☑ In the NICU:

- Use servocontrolled warmer or incubators.
- Use warm IV fluids and blood products, etc.
- Use of plastic tents (cling wrap) and applying cream/oil (like coconut oil) reduces both convection heat loss and insensible water loss. Cream/oil use is restricted to <72 hours duration.
- On discharge from NICU both the abdomen and feet should feel warm normally.
- Placing the newborn in Kangaroo mother care in the NICU reduces neonatal hypothermia significantly.

☑ Postnatal ward:

- Healthy neonates in postnatal wards often develop neonatal hypothermia. This can be prevented by ensuring skin-to-skin care of these neonates regardless of gestation/weight as well as ensuring shared bedding with mother. This is above and beyond the recommendation of 1 hour of postdelivery skin-to-skin care. Most neonatal hypothermia occurs in the first 6 hours of delivery and ensuring skin-to-skin care in these hours can reduce neonatal hypothermia significantly.

☑ Transport:

- A stable infant can be wrapped in warm blanket and cap.
- For a sick infant, transport incubator is the preferred method of transport from delivery room to NICU or intrahospital transfers or from one hospital to another.
- In the absence of transport incubators, a combination of plastic bag + skin-to-skin + cap can be used.
- Phase change material is also utilized for warm transport in many areas across India.
- Kangaroo mother care can be used as an alternative for neonatal transport.

☑ *Rewarming a hypothermic baby:* Warm the room, bed, and use warm blanket and cap to cover the baby, if not already done **(Table 3)**.

TABLE 3

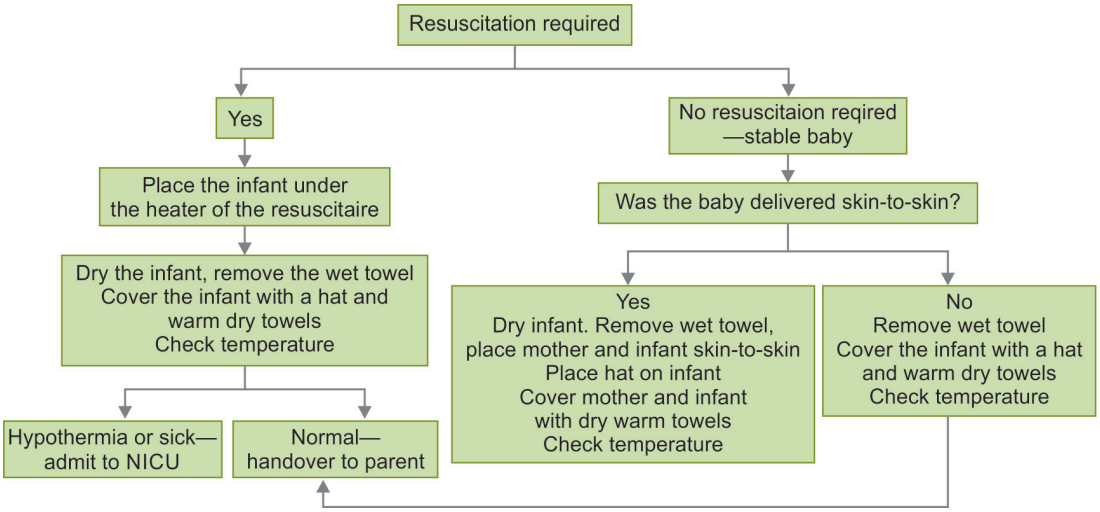
Mild hypothermia	***Moderate hypothermia***	***Severe hypothermia***
☑ Kangaroo (skin-to-skin) care and cover the baby adequately ☑ Heat source can be increased by 0.5° every 30 minutes	☑ Kangaroo (skin-to-skin) care and cover the baby adequately ☑ Rewarm in an incubator or a radiant warmer, if available ☑ Rewarm at a maximum of 0.5° every 30 minutes	☑ Admit in hospital and rewarm in an incubator or a radiant warmer ☑ The temperature is set at 35–36°C and rapidly rewarmed. Once baby's temperature reaches 34°C the rewarming process is slowed down ☑ Supportive management with oxygen and fluids should be started along with appropriate monitoring of vitals and blood sugar

Remove all clothes while rewarming in an open care radiant warmer. Recheck axilla temperature every 30 minutes after each intervention. Consider infection if a baby does not respond adequately to treatment.

☑ *Measurement of toe-core gap:* A difference of >2–3°C between the core and peripheral temperature is abnormal. This gives an early indication of cold stress, hypovolemia, infection, and iatrogenic overheating.

☑ A term baby is being born by normal delivery. How will you maintain baby's temperature? **(Flowchart 1)**

Flowchart 1



☑ A 29-week gestation baby is being born by emergency lower segment cesarean section (LSCS) due to maternal pregnancy-induced hypertension (PIH). How will you maintain the temperature of this baby during the golden hour?

For this baby under 30 weeks gestation, follow the **Flowchart 1** with certain modifications. Place the newborn on radiant heat warmer but do not dry the infant. Instead, place the infant in a plastic bag, leaving the face exposed. Cover the head with a hat. Then transfer to NICU, preferably in a preheated transport incubator.

Further Reading

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